School of Engineering
The University of Warwick

General Assignment Information

Module Code	ES3F1-ES4F3
Module Title	High Performance Embedded Systems Design
Assignment Title	Design Assignment
Assignment Weighting / Credits	50% / 7.5 CATS
Submission Deadline	Wednesday, 12 noon, Week 30

Intended Learning Outcomes (ILOs) Assessed

Intended Learning Outcome(s) (ILOs)	**Criterion**
Communicate effectively the complexity and creativity of a hardware-software co-design implementation on FPGA.	Compose a report summarizing the testing strategies employed to validate your design and how they are connected to the testing strategies discussed in the module. Introduction and Background discussion on baseline design Matrix multiplication labs discussion Design description Code quality, presentation, accordance to report template and references Reflection section
Apply advanced features of FPGA architectures in high performance embedded systems design. Design a hardware accelerator for a complex algorithm by evaluating its parallelism and arithmetic requirements. Apply specialist practical knowledge of hardware design at the register transfer level and understand the principles of high level synthesis.	Create a video processing system for prototyping on the Zybo-Z7 FPGA. Prepare a report detailing the development process and the implemented features of your design. A working video processing module with single-pixel operations A working video processing module with multi-pixel operations Extra features Creative design ideas

- A total mark below 30% (or 40% for Year 4/MSc) indicates that the ILOs have not all been met at threshold level;
- A total mark in the range 30 – 38% (or 40 – 48% for Year 4/MSc) indicates that the ILOs have all been partially met to at least threshold level;
- A total mark of at least 40% (or 50% for Year 4/MSc) indicates that the ILOs have all been met.

Submission Details

The submission details for this assignment are:

- **Deadline:** Wednesday, 12 noon, Week 30.
- **Method:** Online Submission via Tabula
- **Format of submission:** The submission is composed of **two parts**, details for this assignment are:
    1) *A single .pdf file with a meaningful filename that would contain the student number, the module code, and assignment name, for example: "1234567_ES3F1-ES4F3_Lab Report.pdf"*

    Submission length: 10 pages. You will lose **5 marks** per extra page over the limit.

    Formatting instructions: Use a minimum 11-point Arial (or equivalent) font for the text, with 1.5 line spacing and 25 mm margins all round.

    2) *A single .zip file containing (i) video submission and (ii) Vivado project folder with your design. Use a meaningful filename that would contain the student number, the module code, and assignment name, for example: "1234567_ ES3F1-ES4F3_Demonstration.zip"*

    Submission length: Maximum 10-minute video. You will lose **5 marks** per extra minute over the limit.

Note: Submissions should be of an appropriate file size and students are responsible for ensuring that work is uploaded successfully before the deadline. If there are technical issues when submitting online, please contact the Engineering Student Office (eng.eso@warwick.ac.uk).

Guidance and Referencing Style

It is serious Academic Misconduct to pass off the work of others (including peers or AI-based chatbots such as ChatGPT) as your own and you should not permit colleagues to copy from you. Sources must be appropriately and properly acknowledged every time reference is made to another's work, using the Harvard Referencing system. Failure to do so amounts to plagiarism which breaches university regulations and falls short of the Academic Integrity expected in the department and university.

Find out more about the School of Engineering Referencing System here:

https://warwick.ac.uk/fac/sci/eng/eso/undergraduate_students/guidance/handbook/skills/shb-2-04

There are also other types of academic offences including duplication or 'self-plagiarism'. Refer to https://warwick.ac.uk/fac/arts/history/students/undergraduate/assess-plagiarism/ for further details.

Style and Formatting Guide

Submissions are expected to conform to professional standards on style and formatting, and guidance can be found here:

https://warwick.ac.uk/fac/sci/eng/eso/undergraduate_students/guidance/handbook/skills/sh-1-06/

Assignment Feedback

Students will receive group (presentation) and individual (report) feedback on their submission. Feedback on students' assignments will be provided through the Tabula system, together with the release of their awarded marks. The feedback will comprehensively address the strengths of the submission, pinpoint areas for improvement, and suggest actionable steps for students to enhance their future work. Cohort level feedback will be given in term 3.

The Brief

DO NOT TOUCH THE BOARD EXCEPT AT THE SWITCHES

not only for safety, overheating, but also to protect boards from Electro Static Discharge (ESD)

Overview

The assignment for ES4F3/ES3F1 consists of a pair design project building a video processing system on the Zybo-Z7 FPGA development board. You will complete the setup of this platform with the camera by Lab 4 and have the required infrastructure to complete pixel processing operations on a live video stream. You will have access to the FPGA board, camera to work on your own.

You should (i) submit a project report and (ii) do a demonstration session to show and explain your design. The report should include:

- 3 pages discussion on the matrix multiplication experiments from Labs 1–3
- A discussion of the video design setup
- Details about the processing you developed and any additional features added

You are expected to **work in pairs** at your own time, and to **write the report individually**. The assignment is worth 50% of your final grade so please afford the corresponding effort.

You can use the tools at your disposal:

- RTL design in Verilog, replacing the provided colour_change is likely the most straightforward
- Using HLS is not required or worth extra marks in and of itself, but is acceptable
- Xilinx also offers HLS video libraries which you are free to incorporate with correct declaration, but you may find that this is not as straightforward as expected

Up to 40% of the marks will be allocated to the demo (i.e. **achieved working system**) and up to 60% of the marks to the report.

Presentation/Demonstration of practical design

The practical part of the assignment will be assessed by the deadline shown on the cover page. Up to 40% marks will be allocated to this part of the assignment, based on the following breakdown:

- A working video processing module with single-pixel operations: up to 15%
- A working video processing module with multi-pixel operations: up to 15%
- Extra features: up to 5%
- Creative design ideas: up to 5%

You will have to submit **a maximum 10-minute video**, describing how your video processing meets the requirements listed above. You should explain which filters you have implemented and extra features used. Please also include the **Vivado project folder with your design**. Pairs will receive the same mark unless one among the pair contributes more to the work, which

will be assessed through attendance, work in the lab and via teams group monitoring, and may result in deductions.

Written Report Assessment

The written report must be submitted by the deadlines shown on the cover page and **contain maximum 10 pages**.

In this report, you are to describe the design you built. A good report would have the following template structure:

- Introduction to work being shown in the report
  - How the Zynq platform works
  - Overview of HLS design
- Description of Labs 1-3
  - Overview of what is done in each lab
  - Results and discussion of the lab
- Design of the filter:
  - Top view of the project with a figure showing how each module is connected to others.
  - Module design: each module is explained in detail separately
  - Extra features
- Testbench and tests
- Conclusion:
  - Reflection

Start by describing the protocol and interface used for the colour change per pixel, how the design is connected to other modules. How you then improved the baseline design and the work that you have done. Your report should be written in a formal manner and include excerpts of code where this would assist discussion. Screen captures may also be useful. Include all your code in an appendix, clearly stating what each source file is for. **You should use syntax highlighting - do not paste screenshots of code – use an online syntax highlighting tool**. The report should conclude with a reflective section about what you liked and disliked about the project, what you learned or couldn't understand, and suggestions for improvement in the future. Up to 60% marks will be allocated to this part of the assignment, based on the following breakdown:

- Introduction and Background discussion on baseline design: up to 5%
- Matrix multiplication labs discussion: up to 15%
- Design description: up to 25%
- Code quality, presentation, accordance to report template and references: up to 10%
- Reflection section: up to 5%

Marking Scheme

Written Report						
		Introduction and Background discussion on baseline design	**Matrix multiplication labs discussion**	**Design description**	**Code quality, presentation, accordance to report template and references**	**Reflection section**
Class	**Mark**	5%	15%	25%	10%	5%
	85+	Exceptional work of the highest quality, demonstrating excellent knowledge and understanding, analysis. Work of publishable standard. Work which may extend existing debates or interpretations. Needs to have comparison with the state of the art in this field.				
First	**70 - 84**	Excellent introduction, clear background, and objectives. Contains excellent overview of Zynq platform and HLS.	Discussion of Lab 1-3 is excellent, containing an overview of what is done in each lab. Results are shown in a table and later discussed. For lab 2 the report compared different solutions in terms of LUTs, FFs, DSPs and frequency and latency. Lab 3 results are presented in a table, comparing the implementation in hw vs sw, and later discussed correctly.	Description is well written, details with a diagram/figure how the design is structured and all modules are detailed separately. Includes an excellent introduction to the subject and explains what the report contains.	Code is exceptionally well written, contains excellent comments, proper names for signals and wires, excellent indentation, excellent granularity of modules and excellent organization of code in chunks of text. Report follows the template with sections, use correct referencing style, citation of figures, tables and code is correct.	Conclusion section reviews the work presented, includes future works. Conclusion section discusses possible improvements in the design and provides enough reflection on limitations. Reflection section discusses abilities you learned and included those which might be improved.
Upper Second (2-1)	**60 - 69**	Good introduction, clear background, and objectives. Contains good overview of Zynq platform and HLS.	Discussion of Lab 1-3 is good, containing an overview of what is done in each lab. Results are shown in a table and later discussed. For lab 2 the report compared different solutions, but lacks some of the metrics. Lab 3 results are presented in a table, comparing the implementation in hw vs sw, but lacks a discussion or discussion is incomplete.	Description is good, details with a diagram/figure how the design is structured and some of the modules are detailed separately. Includes a good introduction to the subject and explains what the report contains.	Code is well written, contains some comments, excellent indentation, good granularity of modules and good organization of code in chunks of text. Report does not follow one of those: template with sections, use correct referencing style, citation of figures, tables and code is correct.	Conclusion section reviews the work presented, includes future works. Conclusion section discusses possible improvements in the design and provides enough reflection on limitations. Reflection section discusses abilities you learned and included those which might be improved.

Lower Second (2-2)	**50 - 59**	Good introduction but lacks one of the following: background or objectives. Overview of Zynq platform and HLS is incomplete or one is not present.	Some of the discussions of Lab 1-3 are appropriate, containing an overview of what is done in some of the labs. Some results are shown in a table and later discussed. For lab 2 the report compared different solutions but lacks some of the metrics and discussion. Lab 3 results are presented in a table, but incomplete and lacks a discussion or discussion is incomplete.	Description is good but lacks a diagram/figure or an explanation of how the design is structured. Some of the modules are detailed separately. Contains a simple introduction to the subject.	Code lacks some comments, has indentation in some parts, lacks good granularity of modules and/or good organization of code in chunks of text. Report does not follow two of those: template with sections, use correct referencing style, citation of figures, tables and code is correct.	Conclusion section could have been improved, lacks discussion on improvements and/or reflection on limitations. Reflection section could have been improved, includes limited discussion on abilities you learned and those which might be improved.
Third (for ES3F1), FAIL (for ES4F3)	**40 - 49**	Introduction lacks or the following is incomplete: background or objectives. Overview of Zynq platform and HLS is incomplete or not present.	The section discussing lab 1-3 is short, only containing an overview of what is done in each lab or it is incomplete. It lacks information about the aims of the labs, what we are trying to measure and achieve. Labs 1-3 no results are shown neither discussed.	Displayed basic knowledge and understanding of the design, how it is structured and knowledge of relevant subject matter.	Code lacks some comments, lacks indentation, lacks good granularity of modules and/or good organization of code in chunks of text. Report does not have a defined template and order, does not use correct references and/or citation of figures, tables or codes.	Conclusion section is short or does not discuss improvements and/or reflection on limitations. Reflection section does not discuss abilities you learned and/or those which might be improved.
FAIL	**0 - 39**	no accordance to previous points or no submission	no accordance to previous points or no submission	no accordance to previous points or no submission	no accordance to previous points or no submission or don't use syntax highlighting or code is pasted as screenshot	no accordance to previous points or no submission

Demonstration of practical design

		A working video processing module with single-pixel operations	A working video processing module with multi-pixel operations	Extra features	Creative design ideas
Class	**Mark**	15%	15%	5%	5%
	85 +	Exceptional work of the highest quality, demonstrating excellent knowledge and understanding, analysis. Work of publishable standard. Work which may extend existing debates or interpretations. Needs to have comparison with the state of the art in this field.			
First	**70 - 84**	At least three single-pixel operations are shown, with no bug.	At least three multi-pixel operations are shown with implementation using memories for the row-buffers, with no bug.	Extra features are implemented, composed of all of the following: design makes use of board peripherals, software implementations and extra operations on the screen.	Design is innovative, contains multiple features which are creative, for example own animations, easter eggs, design from scratch features, etc
Upper Second (2-1)	**60 - 69**	At least two single-pixel operations are shown, with no bug.	At least two multi-pixel operations are shown with implementation using memories for the row-buffers, with no bugs.	Some extra features are implemented, composed of at least two of the following: design makes use of board peripherals, software implementations and/or extra operations on the screen.	Design does include two creative features
Lower Second (2-2)	**50 - 59**	One single-pixel operation is shown, with no bug.	One multi-pixel operation is shown with implementation using memories for the row-buffers, with no bug.	Some extra features are implemented, composed of at least one of the following: design makes use of board peripherals, software implementations and/or extra operations on the screen.	Design does include one creative feature or more which are partially implemented
Third (for ES3F1), FAIL (for ES4F3)	**40 - 49**	One single-pixel operation is shown, implemented partially or with bugs.	One multi-pixel operation is shown with implementation using memories for the row-buffers partially working or with bugs.	One of the previous extra features is implemented partially or with bugs.	Design does include one creative feature partially implemented
FAIL	**0 - 39**	no accordance to previous points or no submission	no accordance to previous points or no submission	no accordance to previous points or no submission	no accordance to previous points or no submission